Status	Finished
Started	Thursday, 15 October 2026, 4:15 AM
Completed	Thursday, 15 October 2026, 4:19 AM
Duration	3 mins 54 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 100; What are the mid values(corresponding array elements) generated in the first and second iterations?

a. 89 and 94

b. 90 and 100

c. 90 and 99 ✓

d. 94 and 99

Question **2**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

a. 90 and 94

b. 89 and 99

c. 90 and 99 ✓

d. 89 and 94

Question **3**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

a. In the middle of the list

b. In the right sublist ✓

c. In the left sublist

d. At the beginning of the list

Question **4**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

a. Binary search can find multiple instances of the target element
b. Binary search works on unsorted lists
c. Binary search has a lower time complexity on large, sorted lists ✓
d. Binary search does not require dividing the list

Question **5**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

a. When the list is sorted
b. When the element is found in the list ✓
c. When the list contains duplicate elements
d. When the search algorithm finishes

Question **6**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

a. The search stops
b. The search continues in the left or right sublist ✓
c. The search continues from the beginning
d. The list is sorted

Question **7**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It requires the list to be sorted ✓
b. It does not work with negative numbers
c. It can only be applied to large lists
d. It is slower than linear search for small lists

Question **8**

Correct

Mark 1.00 out of 1.00

Which of the following is a type of searching method?

- a. Merge search
- b. Linear search ✓
- c. Quick search
- d. Bubble search

Question **9**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

- a. O(n log n)
- b. O🥐
- c. O(log n) ✓
- d. O(1)

Question **10**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

- a. The list is sorted
- b. The search ends successfully ✓
- c. The search continues in the right sublist
- d. The search continues in the left sublist

Question **11**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

- a. It works on unsorted lists
- b. It compares elements sequentially
- c. It always starts from the beginning of the list
- d. It can be applied only if the list is sorted ✓

Question **12**

Correct

Mark 1.00 out of 1.00

Which method of searching involves sequentially comparing each element until a match is found?

- a. Linear search ✓
- b. Binary search
- c. Hashing
- d. Jump search

Question **13**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

- a. Linear search divides the list into halves.
- b. Linear search is more efficient than binary search on large lists.
- c. Linear search can be applied to both sorted and unsorted lists. ✓
- d. Linear search requires the list to be sorted.

Question **14**

Correct

Mark 1.00 out of 1.00

What type of search would be most appropriate for finding an element in a list that is frequently updated?

- a. Interpolation search
- b. Binary search
- c. Linear search ✓
- d. Hash search

Question **15**

Correct

Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

- a. Hash
- b. Binary ✓
- c. Both (1) & (3)
- d. Linear